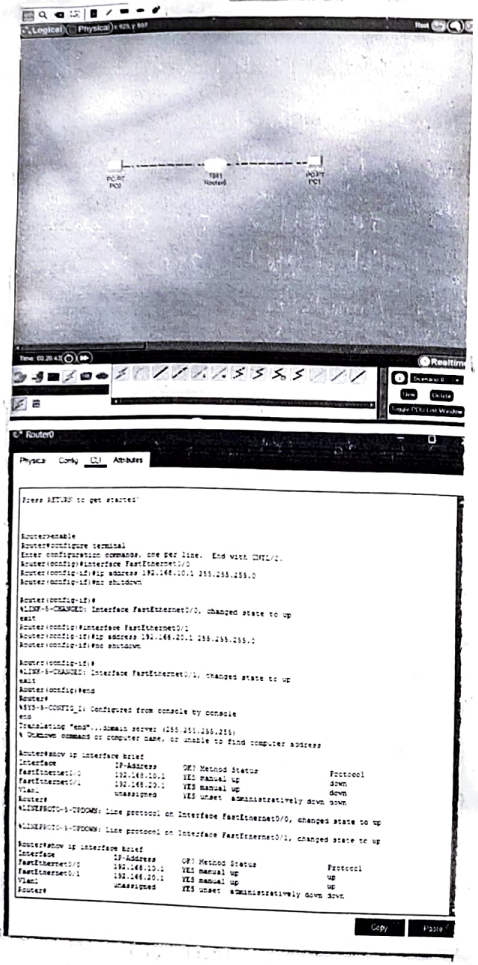


(a)

AIM:

To implement classless IP subnetting Cisco packet Tracer and establish communication between different subnetworks using appropriate IP addresses and subnet masks.

PROCEDURE:

1. open cisco packet Tracer and establish communication between different subnetworks using appropriate IP addresses and subnet masks.

2. Create the topology, in cisco packet Tracer, 2 routers → R_1 & R_2, 2 switches → S_1 & S_2 and 10 Pc's → PC_1 to PC_{10}

3. Connect them by using Copper straight - through for $PC_1 - PC_5$ → S_1, $S_1 \rightarrow R_1$, $PC_6 - PC_{10} \rightarrow S_2$, $S_2 \rightarrow R_2$.

4. Connect Router 1 and Router 2 using a suitable ethernet cable.

5. Use the network address 192.168.1.0/24 and divide it into subnets using the /27 subnet mask (255.255.255.224)

6. Assign suitable IP addresses and subnet masks to the two PCs in the 1st and 2nd subnet.

7. Configure the IP addresses for the correspondig interfaces of R_1 and R_2.

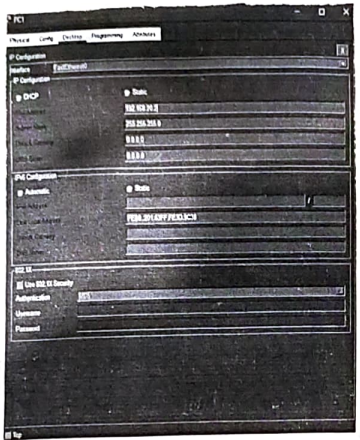
8. Enable all the router interfaces using the no shutdown command.

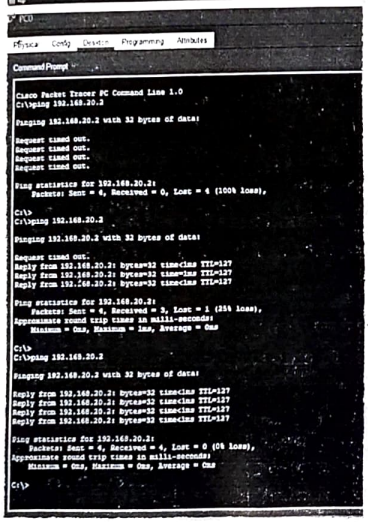
9. Configure the required static routers in R_1 and R_2 to enable communication between the 2 subnets.

10. Verify the configured interfaces using the show ip interface brief command.

11. Test the connectivity between the PC_3 using the ping command.

12. Observe the packet transmission between PC_s in the same subnet and between Pc's in different subnets.





13. Verify that successful ping replies are received, confirming communication between the different subnetworks.

14. Record the observations and results of the subnetting and communication process.

RESULT:

Thus, a simple internetwork using a router was successfully designed and configured in Cisco packet tracer, and communication between the connected networks was successfully established and verified.